$$g_t = \nabla f(\theta_{t-1})$$

$$v_t \leftarrow \beta_2 v_{t-1} + (1-\beta_2) g_t^2$$

$$u_t = g_t / (\sqrt{v_t} + \varepsilon)$$

$$s \leftarrow s + \langle u_t, \theta_t - \theta_0 \rangle$$

$$d \leftarrow \max(d, d_0 + d_coef \cdot s)$$

$$\theta_t \leftarrow \theta - \eta \cdot d \cdot u_t$$